

Expectations from Weekly Exercises – Predictive Modeling

Each weekly exercise should be structured to demonstrate not only technical execution but also a deep understanding of the weekly topic, dataset and the relationships among its variables. Please follow the guidelines below:

1. Structure & Clarity

- **Label Everything:** Clearly label each question and use subsections as needed.
- **Dataset Description:** Start with a brief description of the dataset. For example: “This exercise relates to college data comprising 777 US institutions.”
- **Full Question:** Always include the complete question from the textbook to ensure your answer is perfectly aligned.

2. Code Output and Formatting

- **Key Outputs ONLY:** Don't paste entire code runs or raw data dumps.
- **Neat Tables are King: Extract essential results (e.g., from `summary()` functions) and present them in clean, readable tables.**
- **ISL Style Tables:** For regression, specifically include tables similar to ISL Chapter 3, showing coefficients, standard errors, t-values, and p-values.
- **Selective Code Snippets:** Only include code directly relevant to your analysis or necessary to show how a key result was obtained.

3. Analysis & Insights: Go Beyond the Obvious

This is where you demonstrate mastery. **Move beyond superficial observations to deep, quantitative inference.**

- **Explore Broadly:** Don't just focus on a few variables. Explore at least 80-90% of the dataset's variables to uncover deeper insights and interactions.
- **Quantitative Summaries:** After showing your outputs, provide a clear summary that:
 - **Highlights Key Findings:** Use precise language, e.g., "The regression analysis indicates that the coefficient for variable X is statistically significant ($p\text{-value} < 0.05$), suggesting a strong impact on the dependent variable."
 - **Explains Practical Implications:** What do your findings *mean* in the real world?
 - **Specific, Quantified Observations:** Instead of general statements, provide detailed explanations supported by numbers and plots. For example, "The scatterplot matrix reveals a strong positive relationship between graduation rates and average SAT scores, suggesting that schools with higher standardized test scores tend to have higher graduation rates." (similar to ISL Chapter 2).

- **Support with Data:** Always use quantitative measurements to back up your statements. Include plots and code as supporting evidence.
 - **Hypothesize & Interact:** Develop hypotheses about relationships among *multiple* variables.
 - **Holistic Approach:** Don't isolate variables. Consider interactions, like age, education, and work experience in the "Wage" dataset (ISL example).
 - **Beyond Direct Relationships:** Investigate not only direct predictor-outcome relationships but also potential multicollinearity and interaction effects among various predictors. This reflects the "well-rounded analysis" seen in early ISL chapters.
 - **Precision in Writing:** Your narrative needs to be detailed and specific, directly referencing statistical outputs or visualizations. Ensure correct syntax and terminology, especially when stating hypotheses (null and alternative) and summarizing coefficients.
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In essence, your assignments should not just show *what* you did, but *why* you did it, *what* you found quantitatively, and *what it means* in the context of predictive modeling. This approach demonstrates true mastery and critical thinking.